

Fillerbuster: Unified Generative Scene Completion Model for Casual Captures

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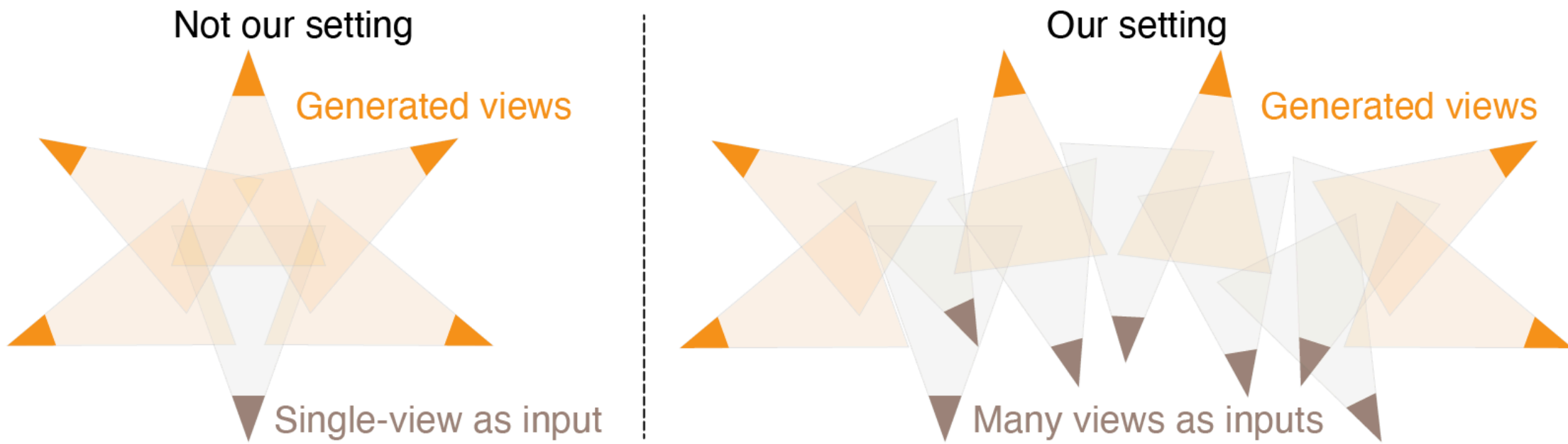
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TL;DR

A unified multi-view diffusion model that jointly completes images and recovers camera poses for casually captured 3D scenes.

- **Unified model** — joint image + pose prediction with latent DiT (650M params)
- **Large context** — condition on 16+ images, generate 24+ consistent views
- **Uncalibrated completion** — recover poses from unposed photos
- **Open source** — model, weights, code for Nerfstudio / Gsplat

Problem Setting



Our setting. Casual captures have hundreds of frames. Single-view methods can't encode the full context.

Scene Completion Results

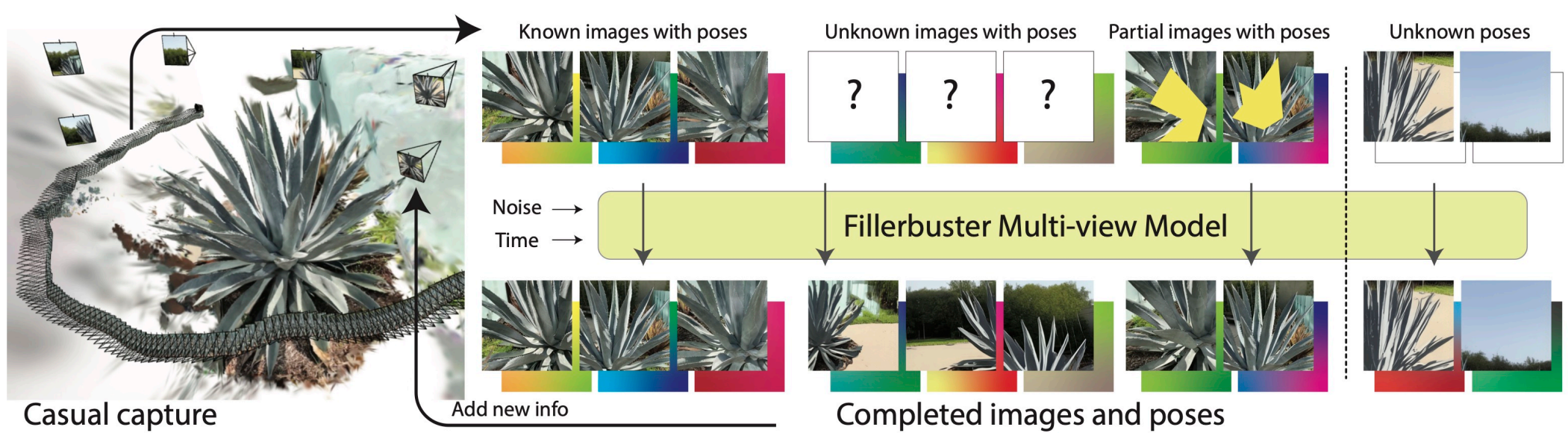


Nerfbusters dataset. 16-40 conditioning images → 24 novel views. Most consistent completions vs. baselines.

Completing Casual Captures



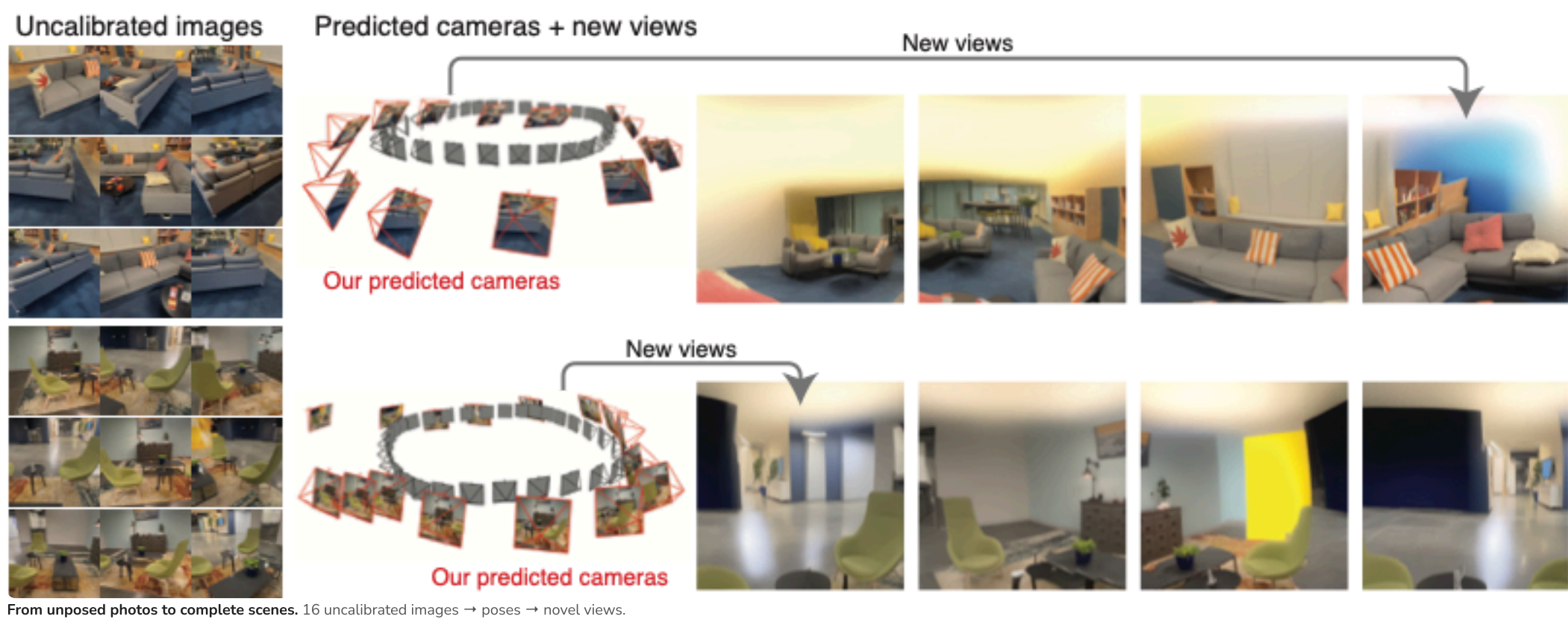
Model Architecture



- Latent DiT (650M params) trained with flow matching on multi-view images + poses
- Separate VAEs for images and raymaps (6-ch ray origins + directions)
- Flexible masking: any mix of known/unknown images and poses
- Index embeddings for variable sequence lengths at inference

$$s_{i,t} = \tilde{z}_{i,t}^I \oplus \mathcal{E}^I(I_i \odot \mathcal{M}_i^I) \oplus \mathcal{D}(\mathcal{M}_i^I) \oplus \tilde{z}_{i,t}^R \oplus \mathcal{E}^R(R_i \odot \mathcal{M}_i^R) \oplus \mathcal{D}(\mathcal{M}_i^R)$$

Uncalibrated Scene Completion



Model Samples



Quantitative Results

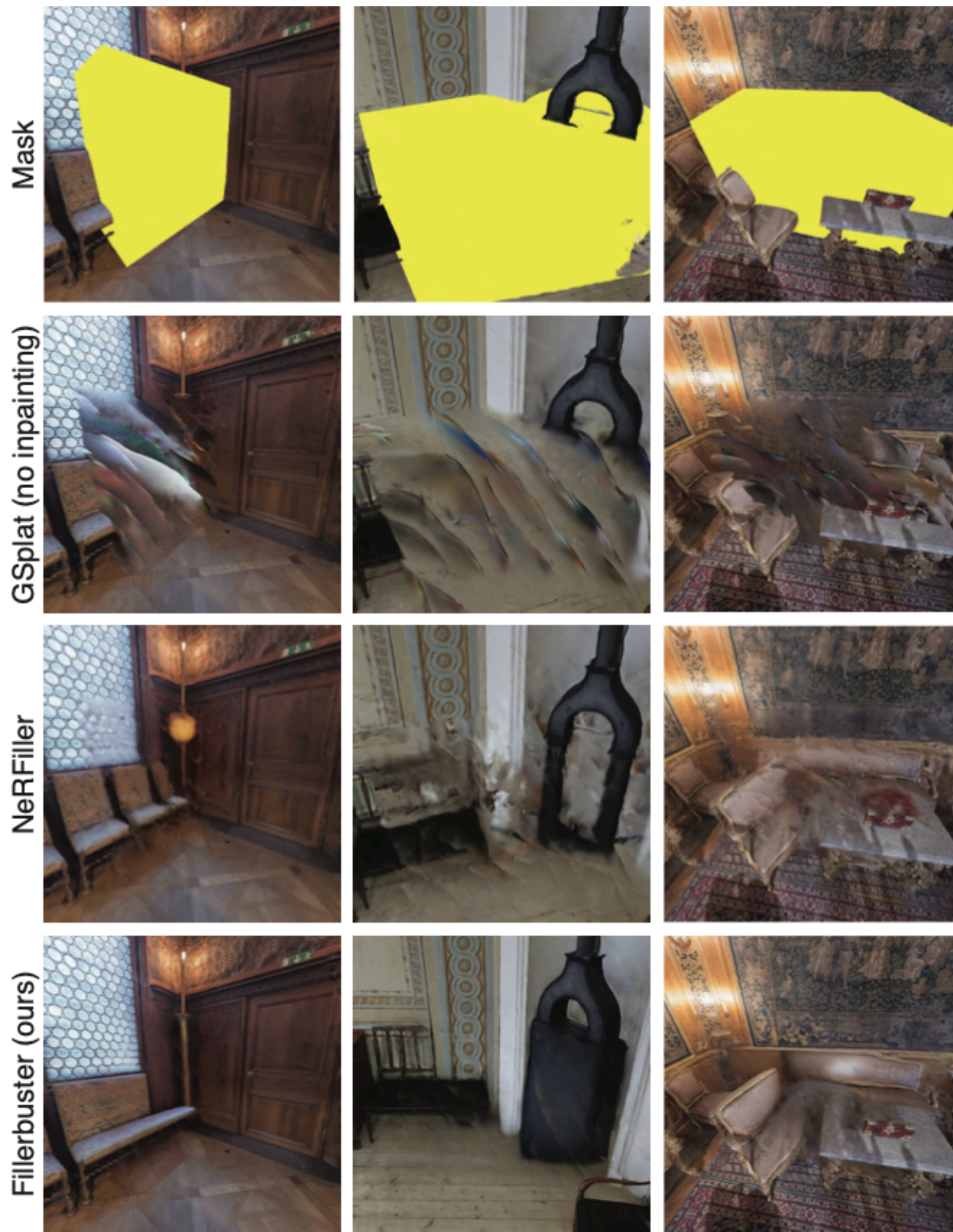
NeRFiler Dataset

Method	PSNR↑	SSIM↑	LPIPS↓
NeRFiler	25.57	0.89	0.182
Fillerbuster	29.60	0.92	0.096

Model Ablations — 16-view

Variant	PSNR↑	SSIM↑	LPIPS↓
no-index-emb	14.06	.450	.400
fixed-index-emb	12.14	.390	.491
no-poses	14.73	.461	.384
Fillerbuster	17.20	.485	.281

NeRFiler Dataset Results



Masked 3D region completion. Higher quality than NeRFiler without depth supervision or iterative updates.

Conclusion & Links

- First unified model for scene completion + pose recovery
- Non-autoregressive joint denoising for consistency
- Open-source: model, weights, training code



Project: ethanweber.me/fillerbuster
Code: github.com/facebookresearch/fillerbuster
arXiv: arxiv.org/abs/2502.05175
Video: youtu.be/_WfzxttMZg